Sardar Vallabhbhai National Institute of Technology, Surat
Department of Computer Science and Engineering
M.Tech (CSE/DS) – I Sem – 2026–27
Computer Vision and Image Processing (CSCS111/CSDS119)

LLM-Assisted CVIP Lab Assignment

Instructions: Each student is assigned a small computer vision task based on an open-source GitHub implementation. The assigned implementation must be studied using an LLM and then modified as specified. The complete solution must not be generated and submitted directly from the LLM.

For the assigned task, the student is expected to:

1. Study and execute the given GitHub implementation.
2. Use an LLM for the specified code-understanding, debugging, or modification task.
3. Make the required modification to the implementation.
4. Test the original and modified implementations.
5. Submit the code, outputs, LLM interaction, and a brief experimental observation.

Roll No.	Problem Statement	LLM Task	What is Expected
P26CS014	**Panorama Creation**	Study the given image-stitching implementation. Ask an LLM to explain how feature detection, feature matching, and image transformation are used in the implementation.	Create a panorama using at least three overlapping images. Modify one parameter or processing step suggested by the LLM and compare the original and modified panorama. Report one situation in which stitching fails. **GitHub:** Source 1 Source 2
P26CS018	**Contour Detection**	Study the contour-detection implementation. Ask an LLM to identify the preprocessing and contour-extraction stages and explain the role of thresholding.	Modify the thresholding/preprocessing stage and test the program on at least three images. Compare the detected contours and identify one case where incorrect contours are obtained. **GitHub:** Source
P26CS026	**Watershed Segmentation**	Study the watershed implementation. Ask an LLM to explain the purpose of foreground, background, and marker regions in the segmentation process.	Apply watershed segmentation to an image containing touching objects. Modify one preprocessing or marker-generation step and compare the segmentation before and after modification. **GitHub:** Source 1 Source 2

Roll No.	Problem Statement	LLM Task	What is Expected
P26DS001	**Face Detection using Haar Cascade**	Study the Haar Cascade face-detection implementation. Ask an LLM to explain the role of the cascade model and the detection parameters used in the program.	Experiment with at least three parameter settings for face detection. Test the detector under changes in scale, illumination, and face orientation. Report successful and failed detections. **GitHub:** Source
P26DS014	**Object Tracking using Optical Flow**	Study the optical-flow implementation. Ask an LLM to identify how motion vectors are calculated and how tracked points are selected or updated.	Run the tracker on a short video. Modify one tracking parameter and compare the tracking performance for slow and fast object movement. Identify one tracking failure. **GitHub:** Source 1 Source 2
P26DS015	**Content-Aware Image Resizing**	Study the seam-carving implementation. Ask an LLM to explain how the energy map and low-energy seam are determined.	Resize the same image using conventional resizing and seam carving. Modify one parameter or step in the seam-selection process and compare the resulting images, particularly around important objects. **GitHub:** Source 1 Source 2
P26DS016	**Object Detection using HOG**	Study the HOG feature implementation. Ask an LLM to explain the purpose of gradient computation, orientation histograms, and block normalization.	Run the detector on suitable images and investigate the effect of changing one HOG parameter. Compare the detection results and discuss one false positive or false negative. **GitHub:** Source 1 Source 2
P26DS017	**Image Super Resolution**	Study the super-resolution implementation. Ask an LLM to explain the difference between the low-resolution input and reconstructed high-resolution output and identify the main model component responsible for upscaling.	Apply the implementation to at least three low-resolution images. Compare the original low-resolution image, conventional interpolation result, and super-resolution output. Report one case where the result is not visually improved. **GitHub:** Source 1 Source 2
P26DS022	**Panorama Quality Analysis**	Use the assigned panorama implementation and ask an LLM to suggest two possible reasons for visible stitching seams or misalignment.	Create panoramas using different amounts of overlap between images. Compare the resulting panoramas and determine how image overlap affects stitching quality. **GitHub:** Source 1 Source 2

Roll No.	Problem Statement	LLM Task	What is Expected
P26DS026	**Watershed Parameter Investigation**	Ask an LLM to identify the step in the watershed implementation that has the greatest influence on separating touching objects and explain why.	Use images containing multiple touching objects. Modify the marker-generation or thresholding step and compare the number and quality of separated regions before and after modification. **GitHub:** Source 1 Source 2
P26DS032	**HOG Detection Parameter Analysis**	Ask an LLM to explain which HOG parameters influence the spatial detail captured by the descriptor and predict how changing one parameter may affect detection.	Run the detector using at least three parameter settings. Compare the detection results and verify whether the LLM's prediction agrees with the experimental results. **GitHub:** Source 1 Source 2

Submission Requirements

1. Source code of the original GitHub implementation used.
2. Modified source code.
3. The LLM prompt(s) used during the task.
4. Original and modified outputs.
5. A brief comparison of the results.
6. One observation supported by the experimental results.
7. At least one failure case or limitation.

Important: The LLM must be used as an assistant for understanding, debugging, parameter selection, or suggesting a small modification. Students must verify the LLM's explanation or suggestion experimentally. Submitting an LLM-generated complete implementation without understanding or modification will not satisfy the assignment.